



Course Name - Linear Algebra and Calculus
Course Code - 25APS0101
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Question Bank

Q Sum and product of the eigen values of the matrix $A = \begin{bmatrix} 2 & 3 & -2 \\ -2 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ is given by respectively

- (a) 5,22 (b) 5,21
(c) 5,40 (d) 5,26

Q The rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$ is

- (a) 3 (b) 2
(c) 1 (d) does not exist

Q If a 5×7 matrix has all its entries equal to 1. Then the rank of the matrix is

- (a) 7 (b) 5
(c) 1 (d) 0

Q. Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & 1 \\ 1 & 0 & 5 \end{bmatrix}$ by using determinant method.

- (a) 0 (b) 1
(c) 2 (d) 3

Q Canonical form of $2xy + 2xz - 2yz$ is

- (a) $x^2 + y^2 - 2z^2$ (b) $x^2 + y^2 + 2z^2$
(c) $x^2 - y^2 - 2z^2$ (d) $x^2 - y^2 + 2z^2$

Q. The eigen values for the matrix $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$ is

- (a) 1, 2 (b) -1, 6
(c) -2, 1 (d) 0, 2

Q For what value of a and b the system of equations $2x+3y+5z=9$, $7x+3y-2z=8$, $2x+3y+az=b$ has a unique solution?

- (a) $a = 5$, $b = 10$ (b) $a \neq 5$, b can be any value
(c) $a \neq 5$, $b = 9$ (d) $a = 5$, $b = 9$

Q Homogeneous system of linear equations

- (a) is always consistent (b) is always inconsistent
(c) has always infinite solution (d) has no solution

Q If $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, then A^8 is

- (a) $A^8 = 5I$ (b) $A^8 = 25I$
(c) $A^8 = 15I$ (d) $A^8 = 625I$

Q. What is the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$ by using Cayley Hamilton theorem.

- (a) $\mu^2 - 7\mu + 1 = 0$ (b) $\mu^2 + 7\mu + 1 = 0$
(c) $\mu^2 - 7\mu - 1 = 0$ (d) $\mu^2 + 7\mu - 1 = 0$

Q The system of linear equations $4x + 2y = 7, 2x + y = 6$ has

- (a) A unique solution (b) Exactly two distinct solutions
(c) Infinite number of solutions (d) No solution

Q For a non-singular matrix A if PAQ is in normal form A^{-1} is equal to

- (a) PQ (b) QP
(c) P-Q (d) P+Q

Q A homogeneous system of linear equations $AX = 0$ in n variables has non-trivial solution if rank A is

- (a) n (b) $>n$
(c) $<n$ (d) $n-1$

Q. The quadratic form of the given equation $2x^2 + 3y^2 + 6xy$ is

- (a) $\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ (b) $\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
(c) $\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ (d) $\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Q The eigen values of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$ are

- (a) 1, 3, 2 (b) 1, 1, 2
(c) 1, 1, 1 (d) 1, 2, 2

Q If 2 & -1 are the eigen values of $A = \begin{bmatrix} 2 & 0 & 0 \\ 5 & 3 & 0 \\ 1 & 3 & -1 \end{bmatrix}$, then the third eigen value is

- (a) -2 (b) 3
(c) 1 (d) -1

Q. The trace of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is

- (a) 5 (b) -2
(c) 1 (d) 4

Q. The Inverse of any rectangular matrix A of order $m \times n$

- (a) Exist (b) Does not exist
(c) Exist if, $|A| \neq 0$ (d) Exist if, $|A| = 0$.

Q A partial differential equation requires

- (a) Exactly one independent variable
(b) More than one dependent variable
(c) Two or more independent variables
(d) Equal number of dependent and independent variables

Q. The equation of normal line to the surface $xyz = a^3$ at the point (x_1, y_1, z_1) is

- (a) $\frac{x-x_1}{y_1 z_1} = \frac{y-y_1}{z_1 x_1} = \frac{z-z_1}{x_1 y_1}$
(b) $-\frac{x-x_1}{y_1 z_1} = \frac{y-y_1}{z_1 x_1} = -\frac{z-z_1}{x_1 y_1}$
(c) $\frac{x-x_1}{y_1 z_1} = -\frac{y-y_1}{z_1 x_1} = -\frac{z-z_1}{x_1 y_1}$
(d) $-\frac{x-x_1}{y_1 z_1} = -\frac{y-y_1}{z_1 x_1} = \frac{z-z_1}{x_1 y_1}$

Q Find the value of a such that the function $f(x) = xe^{ax}$ has a critical point at $x = 3$.

- (a) -0.33 (b) -1
(c) -1.33 (d) 0.33

Q. What is the first order derivative of the given function $z = x^3 + y^3 - 3axy$ w.r.t x is

- (a) $\frac{\partial z}{\partial x} = 3x^2 - 3ay$ (b) $\frac{\partial z}{\partial x} = 3x^2 + 3ay$
(c) $\frac{\partial z}{\partial x} = 3y^3 - 3ay$ (d) $\frac{\partial z}{\partial x} = 3y^3 + 3ay$

Q The degree of the homogeneous function $u = x^3 y^3 / (x^3 + y^3)$ is

- (a) 0 (b) 1
(c) 2 (d) 3

Q If $u = f(y/x)$, then

- (a) $x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} = 0$ (b) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$
(c) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$ (d) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$

Q. What is the degree of $U = \sin^{-1} \frac{x+2y+3z}{x^8+y^8+z^8}$, after the conversion of homogenous function?

- (a) 8 (b) -8
(c) -7 (d) 7

Q If $f(x, y) = x^2 + y^3, x = t^2 + t^3, y = t^3 + t^9$. Find $\frac{df}{dt}$ at $t = 1$.

- (a) 0 (b) 1
(c) -1 (d) 164

Q A function $f(x, y)$ attains minimum at a point (a, b) if

- (a) $rt - s^2 > 0, r < 0$ (b) $rt - s^2 > 0, r > 0$
(c) $rt - s^2 > 0$ (d) $rt - s^2 < 0$

Q If $u = lx + my, v = mx - ly$, then value of $\left(\frac{\partial u}{\partial x}\right)_y \left(\frac{\partial v}{\partial y}\right)_x$ is

- (a) $\frac{1}{l^2 + m^2}$ (b) $\frac{l^2}{l^2 + m^2}$
(c) $\frac{1}{l^2 - m^2}$ (d) $\frac{l^2}{l^2 - m^2}$

Q If u is a homogeneous function of degree n in x and y , then value of $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$ is

- (a) $n(n-1)u$ (b) $n(n+1)u$
(c) nu (d) $-n(n+1)u$

Q. If z is the homogenous function of degree n in x and y variables, then Euler's form is

- (a) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz$ (b) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = (n-1)z$
(c) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2nz$ (d) $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2(n-1)z$

Q. If $z = x^3 + y^3$ be a function of two independent variables x & y . Then

- (a) $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$ (b) $\frac{\partial^2 z}{\partial y \partial x} \neq \frac{\partial^2 z}{\partial x \partial y}$
(c) $\frac{\partial^2 z}{\partial y \partial x} = 2 \frac{\partial^2 z}{\partial x \partial y}$ (d) $\frac{\partial^2 z}{\partial y \partial x} = -2 \frac{\partial^2 z}{\partial x \partial y}$

Q If $f(x, y, z) = 0$, then the value of $\frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x}$ is

- (a) 1 (b) -1
(c) 0 (d) None of these

Q If the function u, v, w of three variables x, y, z are not independent, then Jacobian of u, v, w w. r. t. x, y, z is always equal to

- (a) 1 (b) 0
(c) ∞ (d) $\frac{\partial(x, y, z)}{\partial(u, v, w)}$

Q. If $x = r \cos \theta, y = r \sin \theta$, then

- (a) $\frac{\partial r}{\partial x} = \frac{\cos 2\theta}{x}$ (b) $\frac{\partial r}{\partial x} = \frac{r}{x}$
(c) $\frac{\partial r}{\partial x} = \frac{\cos \theta}{x}$ (d) $\frac{\partial r}{\partial x} = \frac{x}{r}$

Q If $u = x^3 + y^3$, then $\frac{\partial^2 u}{\partial x \partial y}$ is equal to

- (a) -3 (b) 3
(c) 0 (d) $6x + 6y$

Q If u, v are functions of r, s and r, s are functions of x, y , then $J\left(\frac{u,v}{x,y}\right)$ is given by

- (a) $J\left(\frac{u,v}{r,s}\right) J\left(\frac{r,s}{x,y}\right)$ (b) $J\left(\frac{u,v}{r,s}\right) J\left(\frac{x,y}{r,s}\right)$
(c) $-J\left(\frac{u,v}{r,s}\right) J\left(\frac{r,s}{x,y}\right)$ (d) $-J\left(\frac{u,v}{r,s}\right) J\left(\frac{x,y}{r,s}\right)$

Q. If $z = x^2 + y^2$, $x = at^2$, $y = 2at$ is a composite function of t , then the first order total derivative is

- (a) $\frac{dz}{dt} = 2a^2t^3 + 8a^2t$ (b) $\frac{dz}{dt} = 8a^2t^3 + 2a^2t$
(c) $\frac{dz}{dt} = 8a^2t^3 + 4a^2t$ (d) $\frac{dz}{dt} = 4a^2t^3 + 8a^2t$

Q. What is the degree of the function $f(x,y) = 3x^2y + 5y^2x + 4y^3$, according to Euler's theorem on homogeneous function?

- (a) 3 (b) -3
(c) 4 (d) -4

Q Find the rate at which the area of the rectangle is increasing at a given instant when the sides of the rectangle are 4ft and 3ft and are increasing at the rate of 1.5 ft/sec and 0.5 ft/sec. respectively.

- (a) 5.5 sq.ft/sec (b) 6 sq.ft/sec
(c) 6.5 sq.ft/sec (d) 7 sq.ft/sec

Q Linear Taylor series polynomial approximation to function $f(x,y) = x^2y - 2$ about point $(1, -2)$ is

- (a) $y - 4x + 2$ (b) $-4x + y$
(c) $-4x + y$ (d) $-4x + y$

Q If $x + y + z = \log z$, then z_x is:

- (a) $\frac{z}{1-z}$ (b) $\frac{z}{1+z}$
(c) $\frac{1-z}{z}$ (d) $\frac{1+z}{z}$

Q. An implicit function is

- (a) $y - 4x = 3$ (b) $y = \sqrt{x+5}$
(c) $x^2 - y = 3$ (d) $x^5 + y^5 = 5a^3x^2$

Q If $u = x^y$, find $\frac{\partial u}{\partial y}$

- (a) yx^{y-1} (b) x
(c) $x^y \log x$ (d) $x^y \log y$

Q Volume of the ellipsoid $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{4} = 1$ is

- (a) 8π (b) 16π
(c) 4π (d) $\frac{16}{3}\pi$

Q If $u = f(y - z, z - x, x - y)$ then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \dots\dots\dots$

- (a) 1 (b) 0
(c) 2 (d) 3

Q If $f(x, y) = 0$, then $\frac{dy}{dx}$ is equal to

- (a) $\frac{f_x}{f_y}$ (b) $-\frac{f_x}{f_y}$
(c) $\frac{f_y}{f_x}$ (d) $-\frac{f_y}{f_x}$

Q. If u is a homogeneous function of order n , then $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$, both are homogeneous functions of the order:

- (a) n (b) $n - 1$
(c) $n + 1$ (d) $n \pm 1$

Q If $u = \log x + \log y$, then the value of $x u_x + y u_y$ is

- (a) u (b) $2u$
(c) 1 (d) 0

Q. If A is an upper triangular, then $\det(A)$ is

- a. Sum of diagonal entries
b. Product of diagonal entries
c. Always 0
d. None of these

Q. Let A be an $m \times n$ matrix with rank m and B be a $p \times m$ matrix with rank p . What can be the rank of BA ? ($p < m < n$)

- a. m
b. p
c. n
d. $p + m$

Q The value of integral $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$ is

- (a) $\frac{\pi}{32}$ (b) $\frac{3\pi}{32}$
(c) $\frac{5\pi}{32}$ (d) $\frac{7\pi}{32}$

Q The value of the integral $\int_0^2 \int_1^2 \int_0^{yz} xyz \, dx \, dy \, dz$ is:

- (a) 15 (b) $\frac{15}{2}$
(c) $\frac{15}{4}$ (d) $\frac{15}{8}$

Q The triple integral $\iiint dx \, dy \, dz$ over the region R gives:

- (a) volume of region R (b) area of region R
(c) surface area of region R (d) density of region R

Q Beta function $\beta(m, n)$ is defined by

- (a) $\int_0^1 x^m (1 - x)^n dx$ ($m > 0, n > 0$)
(b) $\int_0^1 x^{m-1} (1 - x)^n dx$ ($m > 0, n > 0$)
(c) $\int_0^1 x^m (1 - x)^{n-1} dx$ ($m > 0, n > 0$)

(d) $\int_0^1 x^{m-1}(1-x)^{n-1}dx$ ($m>0, n>0$)

Q Value of $\Gamma(n)\Gamma(1-n)$ is given by

- (a) $n!$ (b) $\frac{\pi}{\sin n\pi}$
(c) $(n-1)!$ (d) $\frac{1}{\sin n\pi}$

Q Change the order of integral $\int_{y=-2}^{y=1} \int_{x=y-2}^{x=-y^2} dx dy$

- A) $\int_{x=-4}^{x=-1} \int_{y=-\sqrt{-x}}^{y=x+2} dy dx + \int_{x=-4}^{x=-1} \int_{y=-\sqrt{-x}}^{y=+\sqrt{-x}} dy dx$
B) $\int_{x=-4}^{x=-1} \int_{y=-\sqrt{-x}}^{y=x+2} dy dx + \int_{x=-1}^{x=0} \int_{y=-\sqrt{-x}}^{y=+\sqrt{-x}} dy dx$
C) $\int_{x=-4}^{x=0} \int_{y=-\sqrt{-x}}^{y=x+2} dy dx + \int_{x=-4}^{x=0} \int_{y=-\sqrt{-x}}^{y=+\sqrt{-x}} dy dx$
D) None of these

Q. The value of integral $\int_0^1 \int_0^x \int_0^{x+y} (x+y+z) dz dy dx$ is

- (a) $\frac{7}{8}$ (b) $\frac{7}{2}$
(c) 0 (d) $\frac{3}{8}$

Q. The value of integral $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$ using spherical coordinates is

- (a) $\frac{\pi^2}{8}$ (b) $\frac{\pi^2}{2}$
(c) $\frac{\pi^2}{6}$ (d) $\frac{\pi}{8}$

Q. The area bounded by the curves $y = x^2$ and $y = x$

- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{6}$ (d) $\frac{1}{4}$

Q. The value of the integral $\int_0^1 (x \log x)^4 dx$ is

- a) $\frac{3!}{5^5}$ b) $\frac{5!}{4^5}$
c) $\frac{4!}{5^5}$ d) $\frac{4!}{5}$

Q. Evaluate the value of the integral $\int_0^1 (\log x)^5 dx$

- a) -120 b) 120
c) 134 d) 124

Q. The function $f(x, y) = x^{\frac{3}{x^2+y^2}}$ is homogeneous of degree

- A) 1 B) 2
- C) 3 D) 4

Q. If $z = \frac{\frac{1}{x^3} + \frac{1}{y^3}}{\frac{1}{x^2} + \frac{1}{y^2}}$, Then the value of $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} =$

- A) $\frac{1}{6}z$
- B) $\frac{1}{3}z$
- C) $\frac{-1}{6}z$
- D) $\frac{1}{2}z$

Q. If z is homogeneous function of degree n in x and y , Then the value of

$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} =$$

- A) nz
B) $n(n-1)z$
C) $n(n+1)z$
D) $\frac{n(n-1)z}{2}$

Q. The asymptote parallel to x-axis for the curve $x^2y - y - x$ is given by

- (A) $y=1$ (B) $y=0$ (C) $x = 0$ (D) $x = 1$

Q. The curve $y = e^x$ approaches the x-axis as $x \rightarrow -\infty$. This is an example of

- (a) A horizontal asymptote
(b) A vertical asymptote
(c) A slant asymptote
(d) No asymptote

Q. The curve $x^3 + y^3 = 3axy$ is symmetric about the line

- A) $y = -x$ (B) $y = x$ (C) $y = x^2$ (D) $y = 0$

Q. In a curve tracing, A double point is classified as a node, if the tangents at this point are

A) real and distinct (B) real and coincident (C) imaginary (D) None of these

Q. In a curve tracing for the function $y^2(2a - x) = x^3$, the double point (0,0) is _____

A) Node (B) Cusp (C) Conjugate Point (D) None of these

Q. Which of the following relation is not true

- (a) $\text{curl}(\text{linear velocity}) = \frac{1}{2} \text{angular velocity}$
- (b) $\text{Curl}(\text{grad}\phi) = 0$
- (c) $\text{Div}(\text{grad}\phi) = 0$
- (d) $\text{Div}(\text{curl}\vec{V}) = 0$

Q. Gamma function is defined by

- (a) proper integral
- (b) improper integral
- (c) indefinite integral
- (d) combination of definite and indefinite

Q. A constant irrotational field is also known as:

- (a) potential field (b) kinetic field
- (c) electrostatic field (d) electromagnetic field.

Q. If $u = x^3 + xy$ and $v = xy$ then value of $\frac{\partial(x,y)}{\partial(u,v)}$ is given by

- (a) $\frac{1}{3x^3}$ (b) $\frac{1}{x^3}$
- (c) $3x^3$ (d) $3x^2$

Q. Value of $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$ is given by

- (a) 1/2 (b) 3/8
- (c) 1/4 (d) 3/5

Q. Beta function representation of the integral $\int_0^1 \frac{x}{\sqrt{1-x^5}} \, dx$ is given by

- (a) $\frac{1}{2} \beta\left(\frac{2}{5}, \frac{1}{2}\right)$ (b) $\frac{1}{5} \beta\left(\frac{2}{3}, \frac{1}{2}\right)$
- (c) $\frac{1}{5} \beta\left(\frac{2}{5}, \frac{1}{2}\right)$ (d) $\frac{1}{2} \beta\left(\frac{1}{5}, \frac{1}{2}\right)$

Q. A function $f(x, y)$ is said to attain maximum value at point (a, b) if

- (a) $rt - s^2 > 0, r > 0$ (b) $rt - s^2 > 0, r < 0$
- (c) $rt - s^2 < 0$ (d) $rt - s^2 = 0$

Q. Value of the integral $\int_0^{\frac{\pi}{2}} \sqrt{\tan x} dx$ is given by

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{\sqrt{2}}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{\sqrt{3}}$

Q. If $u = \log (x^3 + y^3 + z^3 - 3xyz)$ then $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u =$

- (a) $-\frac{9}{(x+y+z)^3}$ (b) $-\frac{9}{(x+y+z)^2}$
(c) $\frac{9}{(x+y+z)^2}$ (d) $\frac{9}{(x^2+y^2+z^2)^3}$

Q. If $f = \frac{x+y}{\sqrt{x}+\sqrt{y}}$, what would be the value of the expression $x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2}$

- (a) $-\frac{1}{4}f$ (b) $\frac{1}{4}f$
(c) $-\frac{1}{2}f$ (d) $\frac{1}{2}f$

Q. Minimum value of $x^2 + y^2 + z^2$ subject to the condition $xyz = a^3$, is given by

- (a) $3a^2$ (b) $-3a^2$
(c) $2a^2$ (d) $3a^3$

Q. Which one of the following is the total derivative of the function $f = \tan^{-1} \frac{y}{x}$

- (a) $\frac{xdy+ydx}{x^2+y^2}$ (b) $\frac{xdy-ydx}{x^2-y^2}$
(c) $\frac{xdy-ydx}{x^2+y^2}$ (d) $\frac{xdx-ydy}{x^2+y^2}$

Q. If $z = f(u, v)$, $u = x \cos y$ and $v = x \sin y$ then $\left(\frac{\partial f}{\partial u}\right)^2 + \left(\frac{\partial f}{\partial v}\right)^2$ is equal to

- (a) $\left(\frac{\partial f}{\partial x}\right)^2 + \frac{1}{x^2} \left(\frac{\partial f}{\partial y}\right)^2$ (b) $\left(\frac{\partial f}{\partial x}\right)^2 + \frac{1}{y^2} \left(\frac{\partial f}{\partial y}\right)^2$
(c) $\frac{1}{x^2} \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2$ (d) $\left(\frac{\partial f}{\partial x}\right)^2 - \frac{1}{y^2} \left(\frac{\partial f}{\partial y}\right)^2$

Q. Volume of the ellipsoid $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{4} = 1$ is

- (a) 8π (b) 16π
(c) 4π (d) $\frac{16}{3}\pi$

Q. What is the value of the integral $\int_0^\infty x^{-\frac{3}{2}}(1 - e^{-x})dx$

(a) $\sqrt{\pi}$

(b) $2\sqrt{\pi}$

(c) $3\sqrt{\pi}$

(d) $\sqrt{2\pi}$

Q. The degree of the homogeneous function $f = \frac{(x+y)^2}{\sqrt{x}+\sqrt{y}}$

(a) 2

(b) $1/2$

(c) $3/2$

(d) 3

Q. Which of the following is the measure of the area bounded by the curves $x^2 = 4y$, $y = 4$ and $xy = 2$

(a) $\frac{28}{3} - 2\log 4$

(b) $\frac{28}{3} + 2\log 4$

(c) $\frac{25}{3} - 2\log 4$

(d) $\frac{28}{3} - 2\log 5$

Q. Value of the integral $\iint_D 4xy - y^2 dx dy$ where D is the rectangle bounded by $x = 1$, $x = 2$, $y = 0$, $y = 3$

(a) 12

(b) 13

(c) 17

(d) 18

Q. If the area of a plane regular region D is given by $\iint_D f(x,y) dx dy$, then value of the function $f(x,y)$ is given by

(a) $4xy - y^2$

(b) $xy - y^2$

(c) $x^2 + y^2$

(d) 1

Q. Which of the following is a vector quantity?

a. mass

b. temperature

c. time

d. force

Q. The value of "a" for which vectors $3i+2j+9k$ and $i+aj+3k$ are parallel is

a. $2/3$

b. $-2/3$

c. $3/2$

d. $-3/2$

Q. Value of the integral $\int_1^2 \int_y^{y^2} \int_0^{\log x} e^z dz dx dy$ is given by

(a) $47/24$

(b) 23

(c) 243

(d) 35

Q. Gradient of a scalar field $\phi(x,y)$ is defined to be

- a. Scalar field with value $\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y}$
- b. Vector field with value $\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j}$
- c. Scalar field with value $\frac{\partial \phi}{\partial x} - \frac{\partial \phi}{\partial y}$
- d. Vector field with value $\frac{\partial \phi}{\partial x} \hat{i} - \frac{\partial \phi}{\partial y} \hat{j}$

Q. Unit tangent vector to the curve $x = 3 \cos t$, $y = -3 \sin t$, $z = -5t$ at any point is

- a. $\frac{-3 \sin t \hat{i} - 3 \cos t \hat{j} - 5 \hat{k}}{\sqrt{34}}$
- b. $\frac{-3 \sin t \hat{i} + 3 \cos t \hat{j} - 5 \hat{k}}{\sqrt{34}}$
- c. $\frac{-3 \sin t \hat{i} - 3 \cos t \hat{j} + 5 \hat{k}}{\sqrt{34}}$
- d. $\frac{3 \sin t \hat{i} - 3 \cos t \hat{j} - 5 \hat{k}}{\sqrt{34}}$

Q. If $\vec{r}$ is a vector function then $\text{div} \vec{r}$ is equal to

- a. 0
- b. 1
- c. 2
- d. 3

Q. The line integral $\int_C x^2 dx + y^2 dy$, C is the square formed by $y = \pm 1$, $x = \pm 1$ is equal to

- a. 0
- b. 1
- c. 4
- d. 2

Q. The value of surface integral $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = (yz)\hat{i} + (xz)\hat{j} + (xy)\hat{k}$ and S is the surface of the sphere having centre at $(0,0,0)$ and radius 1 is equal to

- a. $\frac{3}{8}$
- b. $-\frac{3}{8}$
- c. $\frac{5}{8}$
- d. $-\frac{5}{8}$

Q. If $\vec{a}$ is a constant vector and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\text{curl} (\vec{a} \times \vec{r})$

- a. $2\vec{a}$
- b. $3\vec{a}$
- c. $4\vec{a}$
- d. $\vec{a}$

Q. For $\vec{F} = (x + y + 1)\hat{i} + \hat{j} - (x + y)\hat{k}$, the value of $\vec{F} \cdot \text{Curl} \vec{F}$ is

- a. 1
- b. 2
- c. 3
- d. 0

Q. The directional derivative of $\phi = x^2 + y^2 + z^2$ at the point $(2, 2, 1)$ in the direction of

$2\hat{i} + 2\hat{j} + \hat{k}$ is

a.6 b. 5

c.4 d. 3

Q. Gauss divergence theorem relates

a. Surface and volume integrals

b. Surface and line integrals

c. Line and volume integrals

d. None of these

Q. Stokes theorem relates

a. Surface and line integrals

b. Surface and volume integrals

c. Line and volume integrals

d. None of these

Q. The curve $x^2y + xy + x^2 + y^2 = 4$ intersects the x-axis at the points

(a) $(1,0), (-1,0)$ (b) $(0,0), (0,1)$ (c) $(1,1), (-1,1)$ (d) $(2,0), (-2,0)$

Q. The rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & k \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{bmatrix}$ is 2. Then the value of k is

(a) 1 (b) 2 (c) 3 (d) 4

Q. The region of integration of the double integral $\int_0^3 \int_0^x f(x,y) dy dx$ is a

(a) Rectangle (b) Circle (c) Triangle (d) Square

Q. The divergence of the vector field $x\hat{i} + y\hat{j} + z\hat{k}$ is

(a) 1 (b) 2 (c) 0 (d) 3

Q. The coefficient of x^2y in the Taylor series expansion of the function $e^x \sin y$ about the origin is

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{-1}{4}$ (d) $\frac{-1}{2}$

Q The value of $\beta(5,3)$

- (a) $\frac{\pi}{16}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$

Q The directional derivative of the function xyz at the point $(1,1,1)$ in the direction of the vector $\hat{i}$ is

- (a) -1 (b) $-\frac{1}{3}$ (c) 1 (d) $\frac{1}{3}$

Q The line integral $\int_C x^2 dx + y^2 dy$, where C is the boundary of the region

$$x^2 + y^2 < 4 \text{ equals}$$

- (a) 0 (b) 2 (c) 4π (d) 2π

Q. The value of the double integral $\int_0^2 \int_0^x 3x dy dx$ is a

- (a) 1 (b) 6 (c) 8 (d) 2

Q. The value of $\beta(1,2)$

- (a) 1 (b) $\frac{1}{2}$ (c) 2 (d) 0

Q. The coefficient of x^2 in the Taylor series expansion of the function $e^x \cos y$ about the origin is

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $-\frac{1}{4}$ (d) $-\frac{1}{2}$

Q. The value of $\Gamma\left(\frac{5}{2}\right)$ is

- (a) $\frac{3\sqrt{\pi}}{16}$ (b) $\frac{\pi}{2}$ (c) $\frac{3\sqrt{\pi}}{8}$ (d) $\frac{3\sqrt{\pi}}{4}$

Q. The normal to the surface $x^2 + y^2 + z^2 = 9$ at the point $(1,1,0)$ is

- (a) $\hat{i} + \hat{j} + \hat{k}$ (b) $2\hat{j}$ (c) $2\hat{i} + 2\hat{j} + 2\hat{k}$ (d) $2\hat{i} + 2\hat{j}$

Q. The curl of the vector field $x\hat{i} + y\hat{j} + z\hat{k}$ is

- (a) 1 (b) 2 (c) 0 (d) 3

Q. The product of eigen values of a matrix A is equal to

- (a) A (b) A^T

- (c) $|A|$ (d) Rank A

Q. If λ_i be eigen values of matrix A, then eigen values of A^{-1} is

- (a) $\frac{1}{\lambda_i}$ (b) λ_i^2
(c) $\frac{|A|}{\lambda_i}$ (d) $\frac{1}{\lambda_i^2}$

Q. Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & 1 \\ 1 & 0 & 5 \end{bmatrix}$ by using determinant method.

- (a) 0 (b) 1
(c) 2 (d) 3

Q. The eigen values for the matrix $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$ is

- (a) 1, 2 (b) -1, 6
(c) -2, 1 (d) 0, 2

Q. What is the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$ by using Cayley Hamilton theorem.

- (a) $\mu^2 - 7\mu + 1 = 0$ (b) $\mu^2 + 7\mu + 1 = 0$
(c) $\mu^2 - 7\mu - 1 = 0$ (d) $\mu^2 + 7\mu - 1 = 0$

Q The system of linear equations $4x + 2y = 7, 2x + y = 6$ has

- (a) A unique solution (b) Exactly two distinct solutions
(c) Infinite number of solutions (d) No solution

Q For a non-singular matrix A if PAQ is in normal form A^{-1} is equal to

- (a) PQ (b) QP
(c) P-Q (d) P+Q

Q. Which For $n > 0$, the Gamma function $\Gamma(n + 1)$ is

- (a) $n\Gamma(n)$ (b) $n\Gamma(n + 1)$
(c) $(n + 1)\Gamma(n)$ (d) $n\Gamma(n - 1)$

Q. Every homogeneous system of linear equations is

- a) always consistent
- b) never consistent
- c) may or may not be consistent
- d) none of these

Q. For some scalar λ , if matrix A is such that $(A - \lambda I)$ is singular, then

- a) λ is a characteristic root of A
- b) λ is not a characteristic root of A
- c) λ is zero
- d) λ is eigen value

Q. The sum and product of eigen values of matrix $\begin{bmatrix} 2 & -3 \\ 4 & -2 \end{bmatrix}$ are:

- a) 0, 0
- b) 4, 8
- c) 0, 8
- d) 2, -2

Q. If two or more eigenvalues are equal then corresponding eigen vectors

- a) always linear dependent
- b) always linear independent
- c) may or may not be linear independent
- d) none of these

Q. Cayley Hamilton theorem states that

- a) Every square matrix satisfies its own characteristic equation.
- b) Every matrix satisfies its own characteristic equation.
- c) Every rectangular matrix satisfies its own characteristic equation.
- d) Every square matrix satisfies its own characteristic polynomial.

Q. If z is the homogeneous function of order n , then $x^2 z_{xx} + y^2 z_{yy} + 2xyz_{xy}$ will be given by:

- (a) $n(n - 1)z$
- (b) $(n - 1)z$
- (c) z
- (d) nz

Q. The stationary point of $f(x, y) = xy$ will be

- a) (0, 0)
- b) (1, 1)
- c) (0, 1)
- d) (1, 0)

Q. The rectangular solid of maximum volume that can be inscribed in a sphere is a

- a) Square
- b) circle
- c) Rectangle
- d) cube

Q. The asymptotes parallel to y-axis of the curve $x^2(y^2 - a^2) - b^2y^2 = 0$ is

- (a) $x = \pm a$ (b) $y = \pm a$
(c) Does not exist. (d) None of these.

Q. The value of triple integral $\iiint_V x dV$ where $V\{(x, y, z); 2 \leq x \leq 6, 1 \leq y \leq 2, 0 \leq z \leq 1\}$ is

- (a) $\frac{2}{3}$ (b) 16
(c) 3 (d) 2

Q. Using Gamma Function, the value of $\int_0^\infty x^{7/2} e^{-x} dx$ is

- (a) $\Gamma\left(\frac{5}{2}\right)$ (b) $\Gamma\left(\frac{3}{2}\right)$
(c) $\Gamma\left(\frac{9}{2}\right)$ (d) $\Gamma\left(\frac{1}{2}\right)$

Q. The divergence of a vector point function is a

- a) Vector point function b) Scalar point function
c) Both (a) and (b) d) None of these

Q. The Curl of a vector point function is a

- a) Vector point function b) Scalar point function
b) Both (a) and (b) d) None of these

Q. Any integral which is to be evaluated along a curve is called.....

- a) Line integral b) Surface integral
c) Volume Integral d) None of these

Q. Any integral which is evaluated over a surface is called.....

- a) Line integral b) Surface integral
d) Volume Integral d) None of these

Q. If $\text{Div } \vec{V} = 0$ everywhere in some region R of space, then $\vec{V}$ is called

- a) Solenoidal Vector b) Irrotational vector
b) Volume Integral d) None of these

Q. If $\vec{F} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (x + 2az)\vec{k}$ is solenoidal then the value of a is

- a) $a = 1$ b) $a = 2$

c) $a = -1$

d) $a = -2$

Q. The unit vector normal to the surface $x^4 - 3xyz + z^2 + 1 = 0$ at the point $(1, 1, 1)$ is

a) $\frac{(\vec{i}-3\vec{j}-\vec{k})}{\sqrt{10}}$

b) $\frac{(\vec{i}-3\vec{j}-\vec{k})}{\sqrt{11}}$

c) $\frac{(\vec{i}-3\vec{j}-\vec{k})}{\sqrt{12}}$

d) $\frac{(\vec{i}-3\vec{j}-\vec{k})}{\sqrt{14}}$

Q. The directional derivative of the function $f = xy + yz + zx$ in the direction of vector $\vec{i} + 2\vec{j} + 2\vec{k}$ at the point $(1, 2, 0)$ is

a) $10/3$

b) $10/4$

c) $10/2$

d) 10

Q. The value of $\int y^2 dx - 2x^2 dy$ along the parabola $y = x^2$ from $(0, 0)$ to $(2, 4)$ is

a) $48/5$

b) $-48/5$

c) $48/7$

d) $-48/7$

Q. If $\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, then A^8 is

a) $525I$

b) $630I$

c) $625I$

d) $449I$

Q. The matrix of the quadratic form $x_1^2 + 4x_3^2 + 4x_1x_2 + 10x_1x_3 + 6x_2x_3$ is

a) $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 3 \\ 5 & 3 & 4 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 5 & 5 \\ 2 & 0 & 3 \\ 5 & 3 & 4 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 2 & 3 \\ 5 & 3 & 4 \end{bmatrix}$

d) $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 3 \\ 5 & 3 & 6 \end{bmatrix}$

Q. The product of the eigenvalues of the matrix $\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$ is

a) 3

b) 6

c) 4

d) 1

Q. If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, then $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ is equal to

(a) 0

(b) 1

(c) 2

(d) 4

Q. If $f(x, y, z) = 3yx^2z + 5y^2zx + 4z^4$ is a function of x and y variables, then $\frac{\partial^2 f}{\partial x \partial y}$ is equal to?

a) $2(3xz + 5yz)$

b) $6yxz + 5y^2z$

c) $6yxz - 5y^2z$

d) $6xz - 10yz$

Q. The value of the integral $\int_{-\pi}^{\pi} \sin^5 x \cos^5 x \, dx$ is

a) $\frac{\pi}{5}$

b) 0

c) 1

d) $\frac{\pi}{3}$

Q. if $z = \log(x^2 + y^2)$, Then the value of $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$ is

(a) $\frac{2x}{x^2+y^2}$

b) $\frac{2(x+y)}{x^2+y^2}$

c) $\frac{2y}{x^2+y^2}$

d) $\frac{1}{x^2+y^2}$

Q. Let $a_0 + b_0(x - 1) + b_1(y - 2)$ is Taylor's Series expansion of $f = x^2 + xy + y^2$ about the point (1, 2), the value of a_0 , b_0 and b_1 are

A) $a_0 = 7$, $b_0 = 4$, $b_1 = 4$

B) $a_0 = 0$, $b_0 = 4$, $b_1 = 4$

C) $a_0 = 7$, $b_0 = 4$, $b_1 = 5$

D) $a_0 = 7$, $b_0 = 5$, $b_1 = 5$

Q. The power dissipated in a resistor is given by $P = \frac{E^2}{R}$. The approximate percentage change in P when E is increased by 3% and R is decreased by 2% will be

(a) 18%

(b) 8%

(c) 9%

(d) 12%

Q. The equation of normal line to the surface $xyz = a^3$ at the point (x_1, y_1, z_1) is

- a. $\frac{x-x_1}{y_1 z_1} = \frac{y-y_1}{z_1 x_1} = \frac{z-z_1}{x_1 y_1}$
 b. $-\frac{x-x_1}{y_1 z_1} = \frac{y-y_1}{z_1 x_1} = -\frac{z-z_1}{x_1 y_1}$
 c. $\frac{x-x_1}{y_1 z_1} = -\frac{y-y_1}{z_1 x_1} = -\frac{z-z_1}{x_1 y_1}$
 d. $-\frac{x-x_1}{y_1 z_1} = -\frac{y-y_1}{z_1 x_1} = \frac{z-z_1}{x_1 y_1}$

Q. The equation of the normal plane to the surface $f(x, y, z) = \frac{x^2}{2} - \frac{y^2}{3} - z$ at $(2, 3, -1)$ will be

- a) $\frac{x-2}{2} = \frac{y-3}{-2} = \frac{z+1}{-1}$
 b) $\frac{x+2}{2} = \frac{y+3}{-2} = \frac{z-1}{-1}$
 c) $\frac{x+2}{-2} = \frac{y+3}{2} = \frac{z-1}{1}$
 d) $\frac{x-2}{-2} = \frac{y-3}{2} = \frac{z+1}{1}$

Q. The value of the integral $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dy dx}{1+x^2+y^2}$ is:

- (a) $\frac{\pi}{4} \log(\sqrt{2} + 1)$ (b) $\frac{\pi}{4} \log(\sqrt{2} - 1)$
 (c) $\frac{\pi}{2} \log(\sqrt{2} + 1)$ (d) $\frac{\pi}{2} \log(\sqrt{2} - 1)$

Q. By changing the order of integration the value of the integral $\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} y \, dx dy$ is:

- (a) $\frac{43}{5}$ (b) $\frac{48}{5}$
 (c) $\frac{46}{5}$ (d) $\frac{48}{3}$

Q. The value of the integral $\int_0^3 \int_0^{3-x} \int_0^{3-x-y} dx \, dy \, dz$ is:

- a) $\frac{9}{2}$ b) $\frac{9}{4}$

c) $\frac{3}{2}$

d) $\frac{7}{2}$

Q. The area lying inside the cardioid $r = a(1 + \cos \theta)$ and outside the circle $r = a$ is

(a) $\frac{a}{4}(\pi + 8)$

(b) $\frac{a^2}{4}(\pi + 8)$

(c) $\frac{a^2}{2}(\pi + 8)$

(d) $\frac{a^2}{4}(\pi - 8)$

Q. The integral $\int_0^\infty \sqrt{x}e^{-x^3} dx$ is

a) $\frac{1}{3}$

b) $\frac{\pi}{3}$

c) $\frac{\sqrt{\pi}}{3}$

d) $\frac{\sqrt{\pi}}{2}$

24.

Q. Change the order of integral $\int_{y=-2}^{y=1} \int_{x=y-2}^{x=-y^2} dx dy$

$\int_{x=-4}^{x=-1} \int_{y=-\sqrt{-x}}^{y=x+2} dy dx + \int_{x=-4}^{x=-1} \int_{y=-\sqrt{-x}}^{y=+\sqrt{-x}} dy dx$

E) $\int_{x=-4}^{x=-1} \int_{y=-\sqrt{-x}}^{y=x+2} dy dx + \int_{x=-1}^{x=0} \int_{y=-\sqrt{-x}}^{y=+\sqrt{-x}} dy dx$

F) $\int_{x=-4}^{x=0} \int_{y=-\sqrt{-x}}^{y=x+2} dy dx + \int_{x=-4}^{x=0} \int_{y=-\sqrt{-x}}^{y=+\sqrt{-x}} dy dx$

G) None of these

Q. Find the work done by the force field $F(x, y) = x\hat{i} - y\hat{j}$ in moving a particle in counter clock wise direction around the circle of radius 1 centered at the origin

a) $\frac{3}{4}\pi$

b) π

c) $\frac{\pi}{2}$

d) 2π

Q. The value of $\iiint \nabla \cdot \vec{F} dv$ if $\vec{F} = (x^2\hat{i} + y^2\hat{j} + z^2\hat{k})$ and V is the volume of the region enclosed by cube $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

(a) 4

(b) 3

(c) 1

(d) 2

d) $a = 4, b = 2, c = -1$

d) $\frac{\hat{i}+3\hat{j}+2\hat{k}}{\sqrt{126}}$

d) $a = 4, b = 2, c = 2$

d) $\iiint_V M \, dx dy dz$

Q. If $z = \log(x^2 + y^2)$, Then the value of $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$ is

$\frac{2x}{x^2+y^2}$

(b) $\frac{2(x+y)}{x^2+y^2}$

(c) $\frac{2y}{x^2+y^2}$

(d) $\frac{1}{x^2+y^2}$

Q. The equation of normal to the surface $z^2 = 4(1 + x^2 + y^2)$ at (2, 2, 6) is

(A) $\frac{x-2}{1} = \frac{y-2}{2} = \frac{z-6}{3}$

(B) $\frac{x-2}{4} = \frac{y-2}{4} = \frac{z-6}{-12}$

(C) $\frac{x-2}{4} = \frac{y-2}{4} = \frac{z-6}{3}$

(D) $\frac{x-2}{4} = \frac{y-2}{4} = \frac{z-6}{-3}$

Q. Given $f(x, y) = x^3 + y^3 - 3axy$ then the value of the partial derivative f_{xy} is?

a) 3

b) $3a$

c) $-3a$

d) 0

Q. If z is homogeneous function of degree n in x and y , Then the value of

$$x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} =$$

A) nz

B) $n(n-1)z$

C) $n(n+1)z$

D) $\frac{n(n-1)z}{2}$

Q. The percentage error in the area of a rectangle when an error of $+1\%$ is made in its length and breadth is

(A) 2%

(B) 1%

(C) 3%

(D) 4%

Q. If $f(x, y) = x^3 - 2y^2 + xy = 0$ then $\frac{dy}{dx}$ at the point (1, 1) is

(A) $\frac{3}{4}$

(B) $\frac{4}{3}$

(C) $-\frac{4}{3}$

(D) $-\frac{3}{4}$